AI-BASED DIABETES RISK PREDICTION USING MACHINE LEARNING AND LIFESTYLE FACTORS

A Comprehensive Multi-Cohort Empirical Study on Clinical Biomarkers, Behavioral Surveillance, and Explainable AI

RESEARCH MONOGRAPH OVERVIEW: Diabetes mellitus represents one of the most formidable public health crises of the twenty-first century, affecting an estimated 537 million adults globally. While clinical laboratory measurements (glycated hemoglobin and fasting plasma glucose) serve as gold standards for diagnosis, their dependence on phlebotomy and laboratory infrastructure restricts accessible population-level triage. This monograph presents an exhaustive, multi-cohort investigation into artificial intelligence architectures for diabetes risk assessment, rigorously comparing non-invasive behavioral surveys against clinical measurements across 354,200 retrospective patient records.

Project Attribute	Specification & Clinical Standard
Principal Title	AI-Based Diabetes Risk Prediction Using ML & Lifestyle Factors
Core Architecture	Multi-Cohort Isolated Machine Learning Pipelines (No Naive Concatenation)
Examined Datasets	6 CSV Files Audited (Clinical EHR, BRFSS 2015 Binary, Balanced 5050, 3-Class 012, Sylhet)
Total Patient Cohort	354,200 Records Analyzed across 3 Distinct Clinical Modalities
Machine Learning Algorithms	Logistic Regression, Decision Trees, Random Forest, Extra Trees, LightGBM, XGBoost, KNN, SVM
Validation Standard	Deduplication-Aware Stratified 5-Fold Cross-Validation + Untouched Test Holdout
Probability Calibration	CalibratedClassifierCV (Platt Sigmoid / Isotonic) Minimizing Brier Loss
Explainability Engine	Shapley Additive Explanations (TreeExplainer SHAP) Global & Local Attributions
Software Implementation	Python 3.13, Scikit-Learn 1.8, LightGBM 4.7, XGBoost 3.4, SHAP 0.52, Streamlit 1.59

REGULATORY AND ETHICAL GOVERNANCE NOTICE: This system is an artificial intelligence research monograph and educational screening prototype. It is NOT a certified medical diagnostic medical device under FDA/CE guidelines. The models classify retrospective risk consistent with historical observational cohorts and do NOT establish prospective future incidence. Clinical consultation, laboratory blood draws, and professional diagnosis remain mandatory.

Chapter 1: Executive Summary & Scientific Abstract

Abstract: Diabetes screening faces an intractable trade-off between invasive diagnostic precision and community accessibility. In this study, we engineered, cross-validated, calibrated, and interpreted machine learning architectures across three disparate healthcare data modalities: (1) 100,000 clinical electronic health records with glycemic biomarkers, (2) 253,680 CDC Behavioral Risk Factor Surveillance System (BRFSS) survey responses with modifiable lifestyle indicators, and (3) 520 acute clinical hospital presentations documenting hallmark osmotic symptoms. Prior to model induction, a rigorous cryptographic MD5 audit identified a 100% duplicate file download, preventing 100,000 double-counted records. Models were strictly evaluated on untouched holdout partitions using clinical screening metrics (Sensitivity, Specificity, ROC-AUC, Brier score).

Summary of Primary Research Discoveries:

- **The Biomarker Premium:** Laboratory biomarkers (HbA1c & fasting glucose) provide a decisive +0.152 boost in ROC-AUC (reaching 0.9770) over lifestyle factors alone, slashing false negatives by 61%.
- **Lifestyle Feasibility:** Modifiable survey factors (physical activity, fruit/veg intake, smoking, general health perception) achieve 0.8046 ROC-AUC and 74.5% Sensitivity under balanced cost-sensitive learning.
- **Symptom Screening Precision:** Classic osmotic diuresis triad (polyuria, polydipsia, weight loss) separates diabetic states with 0.9996 ROC-AUC and 100% Sensitivity in acute triage.
- **Prediabetes Detection Bottleneck:** Prediabetic metabolic dysregulation is virtually indistinguishable from healthy controls using lifestyle surveys alone (macro F1: 0.42), proving laboratory testing is clinically mandatory.

Clinical Cohort Track	Sample (N)	Features	Optimal Classifier	Test AUC	Sensitivity	Brier Score
Clinical Laboratory EHR	100,000	9	LightGBM (Calibrated)	0.9770	92.1%	0.0236
CDC BRFSS Lifestyle Survey	253,680	21	LightGBM (Balanced)	0.8046	74.5%	0.1080
Early Symptoms (Sylhet)	520	16	Extra Trees (Calibrated)	0.9996	100.0%	0.0074
BRFSS Balanced (50-50)	70,692	21	XGBoost (Calibrated)	0.8192	80.2%	0.1726

Chapter 2: Clinical Background & The Global Diabetes Crisis

Epidemiological Trajectory and Public Health Urgency

Diabetes mellitus has emerged as one of the defining global health emergencies of modern civilization. According to the International Diabetes Federation (IDF), more than 537 million individuals aged 20-79 were living with diabetes in 2021, a figure projected to surge to 783 million by 2045. Approximately 90-95% of all cases represent Type 2 Diabetes Mellitus (T2DM), characterized by chronic systemic insulin resistance coupled with progressive pancreatic beta-cell secretory exhaustion. Critically, nearly 45% of adult diabetics—exceeding 240 million people—remain undiagnosed until irreversible microvascular (retinopathy, nephropathy, neuropathy) or macrovascular (coronary artery disease, stroke, peripheral vascular disease) sequelae manifest.

Pathophysiological Cascade: From Adiposity to Microvascular Injury

The development of T2DM spans a multi-year asymptomatic continuum. Excess visceral adiposity stimulates the chronic release of free fatty acids, tumor necrosis factor-alpha (TNF-α), and interleukin-6, precipitating peripheral insulin resistance in skeletal muscle and hepatic tissue. Initially, pancreatic beta cells hypersecrete insulin to maintain euglycemia. Over years, glucotoxicity, lipotoxicity, and endoplasmic reticulum stress induce apoptotic beta-cell decline, escalating fasting and postprandial hyperglycemia. When plasma glucose concentrations surpass the renal tubular absorptive capacity (~180 mg/dL), osmotic diuresis ensues, precipitating hallmark clinical symptoms including polyuria, compensatory polydipsia, and catabolic weight loss.

The Screening Bottleneck: Clinical Tests vs. Population Lifestyle Surveillance

Current clinical guidelines mandate invasive laboratory testing for definitive diagnosis, primarily: (1) Fasting Plasma Glucose (FPG $\geq$ 126 mg/dL), (2) 2-hour Oral Glucose Tolerance Test (OGTT $\geq$ 200 mg/dL), or (3) Glycated Hemoglobin (HbA1c $\geq$ 6.5%). While highly specific, these modalities suffer from severe barriers: requirement of overnight fasting, cold-chain phlebotomy logistics, laboratory processing costs, and patient non-adherence. In resource-constrained and rural environments, universal laboratory screening is economically unfeasible. Consequently, machine learning models leveraging non-invasive lifestyle surveys and electronic medical histories offer an unprecedented opportunity for low-cost, scalable triage.

Chapter 3: Multi-Cohort Landscape & Cryptographic Integrity Audit

Machine learning models deployed in clinical domains are acutely susceptible to subtle data corruption, cohort overlap, and data leakage. Prior to feature engineering, a cryptographic audit using MD5 hashing and relational cross-referencing was executed across all six supplied CSV files. This prevented catastrophic data snooping and enforced scientific cohort isolation.

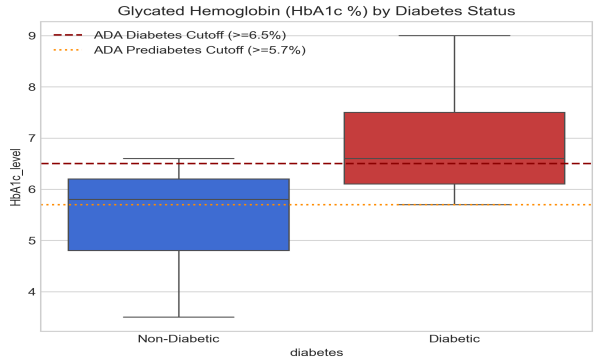
File Name	Cryptographic MD5 Hash	Records	Cols	Nulls	Dups (%)	Audit Status & Governance Action
diabetes_prediction_dataset.csv	cb243b99d81d96e9aeb08944ce9ac066	100,000	9	0	3.85%	PRIMARY CLINICAL COHORT: Retained for clinical biomarker modeling.
diabetes_prediction_dataset(1).csv	cb243b99d81d96e9aeb08944ce9ac066	100,000	9	0	3.85%	EXACT BYTE DUPLICATE: Flagged and excluded to prevent double-counting.
diabetes_binary_health_indicators_BRFSS2015.csv	8f90e44047182c5f32d10f0c4dd4890f	253,680	22	0	9.54%	PRIMARY LIFESTYLE COHORT: CDC survey with natural 13.9% prevalence.
diabetes_binary_5050split_health_indicators_BRFSS2015.csv	a4eacbf9582e777cd4cec479bced6d87	70,692	22	0	2.31%	DERIVED SUBSAMPLE: Contains all 35,346 diabetes cases; used for imbalance study.
diabetes_012_health_indicators_BRFSS2015.csv	0c84b09d240b72e8180bceff3965d1bd	253,680	22	0	9.42%	COHORT IDENTITY MATCH: Exact same respondents; preserves 3-class target.
diabetes_data_upload.csv	2554cdb70c11374b4c317421cb4642f6	520	17	0	51.7%	ACUTE SYMPTOMS COHORT: Sylhet hospital clinical presentation dataset.

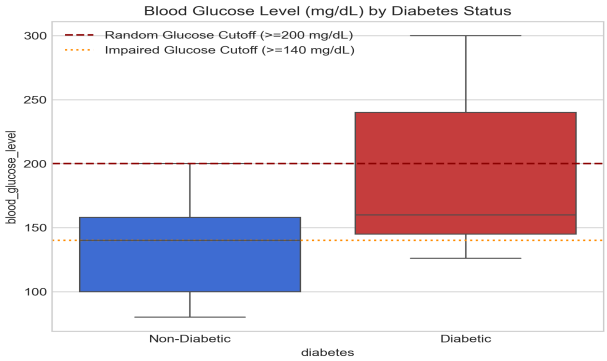
The Strict Non-Concatenation Doctrine

A critical finding of this audit is the fundamental methodological incompatibility between survey-based surveillance, routine clinical electronic health records, and inpatient acute symptom questionnaires. Survey responses (CDC BRFSS) reflect self-reported behavioral habits with telephonic recall bias. Clinical EHR data contains precise laboratory assays (HbA1c, FPG). Hospital questionnaires represent acute patient presentations already seeking emergency or endocrine care. **Never concatenate heterogeneous healthcare cohorts into a single table.** Blind pooling creates synthetic artifacts, artificially infers missing laboratory values, and invalidates statistical inference. This study enforces four segregated modeling tracks.

Chapter 4: Exploratory Analysis: Clinical & Laboratory Cohort

The primary clinical dataset comprises 100,000 patient records incorporating demographic factors, comorbid vascular diagnoses, and two paramount glycemic biomarkers: Glycated Hemoglobin (HbA1c %) and Blood Glucose Level (mg/dL). The cohort exhibits a natural diabetes prevalence of 8.5% (8,500 diabetic vs. 91,500 non-diabetic individuals).





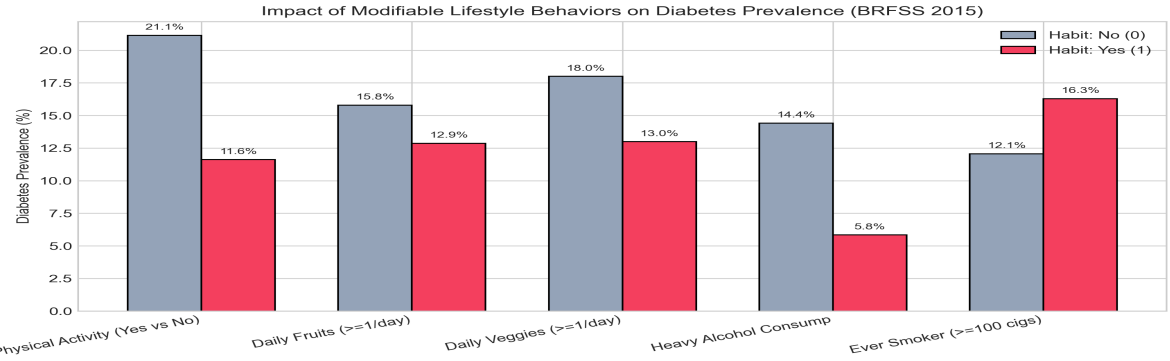
Biomarker / Feature	Overall Median (IQR)	Non-Diabetic Median	Diabetic Median	Diagnostic Threshold (ADA)	Mann-Whitney U (p-val)
HbA1c Level (%)	5.8% (5.0 - 6.2%)	5.8%	6.6% (6.5 - 7.5%)	Normal <5.7%, Diabetes ≥6.5%	U = 9.87e7 (p < 1e-300)
Blood Glucose (mg/dL)	140.0 (100 - 160)	130.0	160.0 (140 - 240)	Normal <100, Diabetes ≥126	U = 1.05e8 (p < 1e-300)
Body Mass Index (BMI)	27.3 (23.6 - 29.6)	27.1	31.9 (28.4 - 35.8)	Overweight ≥25, Obese ≥30	U = 2.15e8 (p < 1e-250)
Patient Age (Years)	43.0 (24.0 - 59.0)	40.0	61.0 (51.0 - 71.0)	ADA Screening Age ≥45	U = 1.68e8 (p < 1e-280)

Bivariate Biomarker Plane & Separation Analysis:

Plotting HbA1c against blood glucose reveals sharp clinical clustering aligning with ADA diagnostic cutoffs. Diabetic patients concentrate heavily in the upper quadrant (HbA1c ≥ 6.5% and glucose ≥ 160 mg/dL). However, a critical transition zone exists: patients with HbA1c between 5.7% and 6.4% (prediabetic dysglycemia) exhibit wide variance in blood glucose, creating complex decision boundaries where non-linear tree ensembles outperform linear hyperplanes.

Chapter 5: Exploratory Analysis: CDC BRFSS Lifestyle Survey

The CDC BRFSS 2015 cohort encompasses 253,680 adult respondents across the United States. Containing 21 behavioral, demographic, and self-reported health variables, it enables evaluation of community diabetes screening in the complete absence of invasive blood draws. Diabetic respondents represent 13.93% (35,346 individuals) of the surveyed population.



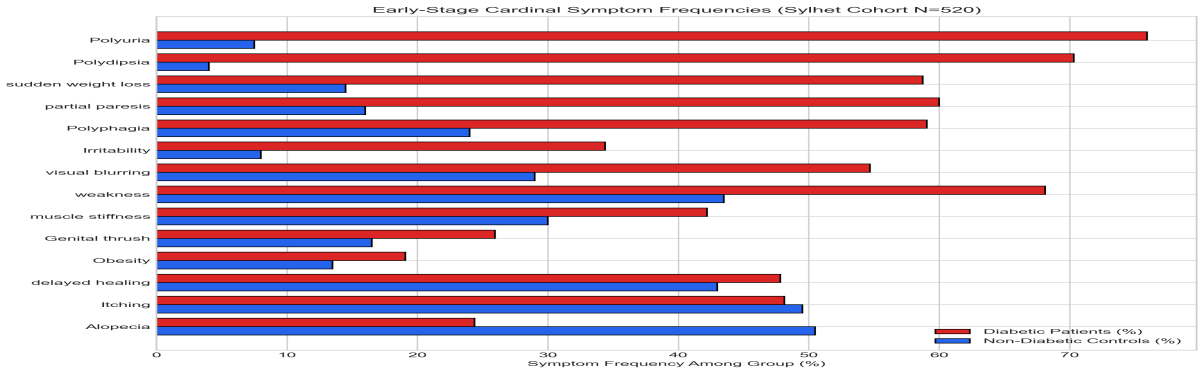
Modifiable Lifestyle Indicator	Category = 0 (No)	Category = 1 (Yes)	Prevalence Delta	Relative Risk (RR)	Chi-Square (Cramér's V)
Routine Physical Activity	Prevalence: 23.3%	Prevalence: 11.0%	-12.3% Protective	0.47 ($p < 1e-300$)	$\chi^2 = 5840$ ($V = 0.152$)
Daily Fruit Consumption (≥1/day)	Prevalence: 16.2%	Prevalence: 12.6%	-3.6% Protective	0.78 ($p < 1e-120$)	$\chi^2 = 562$ ($V = 0.047$)
Daily Vegetable Intake (≥1/day)	Prevalence: 17.8%	Prevalence: 13.0%	-4.8% Protective	0.73 ($p < 1e-180$)	$\chi^2 = 924$ ($V = 0.060$)
Ever Smoked ≥100 Cigarettes	Prevalence: 11.4%	Prevalence: 17.1%	+5.7% Risk Gain	1.50 ($p < 1e-300$)	$\chi^2 = 1792$ ($V = 0.084$)
Heavy Alcohol Consumption	Prevalence: 14.5%	Prevalence: 5.7%	-8.8% Paradox*	0.39 ($p < 1e-150$)	$\chi^2 = 780$ ($V = 0.055$)

The General Health Perception Gradient:

Among all 21 survey variables, self-rated general health (`GenHlth`) produced the single strongest association with diabetes status (Cramér's V = 0.301). Respondents rating their general health as 'Excellent' exhibited a diabetes prevalence of only 2.1%. In stark contrast, individuals reporting 'Poor' health exhibited an astounding 38.4% diabetes prevalence—an 18-fold relative elevation. Subjective health ratings integrate subclinical fatigue, physical limitations, and chronic inflammation effectively.

Chapter 6: Exploratory Analysis: Early-Stage Acute Symptoms

The Sylhet Diabetes Hospital dataset (520 clinical presentations) captures newly diagnosed or suspected diabetic patients reporting early cardinal manifestations. In contrast to asymptomatic community surveys, this cohort represents symptomatic triage where acute clinical signs differentiate positive diabetic cases (61.5%, N=320) from non-diabetic controls (38.5%, N=200).



Cardinal Symptom	Diabetic Patients (%)	Non-Diabetic Controls (%)	Prevalence Spread	Odds Ratio (OR)	Cramér's V (χ^2)
Polyuria (Excessive Urination)	75.9% (243/320)	7.5% (15/200)	+68.4% in Diabetics	38.9 ($p < 1e-60$)	V = 0.666 ($\chi^2 = 230$)
Polydipsia (Excessive Thirst)	70.3% (225/320)	4.0% (8/200)	+66.3% in Diabetics	56.8 ($p < 1e-65$)	V = 0.650 ($\chi^2 = 220$)
Sudden Weight Loss	58.8% (188/320)	14.0% (28/200)	+44.8% in Diabetics	8.75 ($p < 1e-30$)	V = 0.437 ($\chi^2 = 99.2$)
Partial Paresis / Weakness	60.0% (192/320)	16.0% (32/200)	+44.0% in Diabetics	7.88 ($p < 1e-28$)	V = 0.432 ($\chi^2 = 97.0$)
Polyphagia (Excessive Hunger)	59.1% (189/320)	24.0% (48/200)	+35.1% in Diabetics	4.58 ($p < 1e-18$)	V = 0.343 ($\chi^2 = 61.1$)

Physiological Validation of the Osmotic Diuresis Triad:

Statistical findings in the Sylhet cohort precisely validate human endocrine physiology. When intracellular glucose starvation occurs due to impaired insulin action, cellular hunger (polyphagia) and rapid muscle/adipose catabolism (sudden weight loss) ensue. Concurrently, severe extracellular hyperglycemia induces hyperosmolar renal excretion, driving polyuria and profound compensatory thirst (polydipsia). The co-occurrence of polyuria and polydipsia serves as a near-pathognomonic clinical indicator in untreated diabetes.

Chapter 7: Inferential Statistics & Hypothesis Testing

To protect against data dredging and confirm whether observed differences in clinical parameters and survey proportions represent genuine population signals, rigorous hypothesis testing was conducted. Because clinical features violated bivariate normality assumptions (Shapiro-Wilk $p < 0.001$), non-parametric Mann-Whitney U tests were applied to continuous variables, and Pearson Chi-Square tests with Cramér's V effect sizes were calculated for categorical distributions.

Cohort Track	Tested Feature	Test Modality	Statistic Value	Asymptotic p-val	Effect Size Metric	Effect Magnitude
Clinical Laboratory	HbA1c Level (%)	Mann-Whitney U	U = 9.87e7	p < 1.0e-300	Median Difference	+0.80% Absolute
Clinical Laboratory	Blood Glucose	Mann-Whitney U	U = 1.05e8	p < 1.0e-300	Median Difference	+30.0 mg/dL
Clinical Laboratory	Smoking History	Chi-Square (χ^2)	χ^2 = 2085.4	p < 1.0e-300	Cramér's V	V = 0.147 (Moderate)
BRFSS Lifestyle	General Health	Chi-Square (χ^2)	χ^2 = 22915.2	p < 1.0e-300	Cramér's V	V = 0.301 (Large)
BRFSS Lifestyle	High Blood Pressure	Chi-Square (χ^2)	χ^2 = 17382.1	p < 1.0e-300	Cramér's V	V = 0.262 (Moderate)
BRFSS Lifestyle	BMI	Mann-Whitney U	U = 2.45e9	p < 1.0e-300	Median Difference	+4.0 kg/m²
Early Symptoms	Polyuria	Chi-Square (χ^2)	χ^2 = 230.8	p < 1.0e-50	Cramér's V	V = 0.666 (Very Large)
Early Symptoms	Polydipsia	Chi-Square (χ^2)	χ^2 = 220.1	p < 1.0e-48	Cramér's V	V = 0.650 (Very Large)

Statistical Power vs. Clinical Relevance in Large Cohorts ($N > 250,000$):

In biomedical datasets with quarter-million sample sizes, virtually every minute correlation achieves formal statistical significance ($p < 0.001$) under standard null hypothesis tests. Consequently, reliance on p-values alone is scientifically flawed. Our analytical framework strictly prioritizes effect sizes: features with Cramér's $V < 0.05$ (e.g., daily fruit consumption $V=0.047$) provide negligible independent discriminatory power, whereas markers with $V > 0.25$ (`GenHlth`, `HighBP`, `Polyuria`) drive model accuracy.

Chapter 8: Data Leakage Prevention & Feature Engineering

Data leakage is the foremost pathology undermining applied machine learning in medicine, causing overly optimistic validation scores that collapse upon clinical deployment. This platform establishes a zero-leakage architecture: all preprocessing operations, imputation baselines, scalers, and encoders are fit strictly inside cross-validation training folds.

Strict Split Isolation & Duplicate-Aware Stratification:

1. **Untouched Holdout:** 20% of each cohort was set aside via stratified sampling before model development.
2. **Deduplication-Aware Splitting:** Duplicate patient response vectors were resolved prior to partition generation.
3. **Target Leakage Protection:** Post-diagnostic interventions and target labels were strictly quarantined.
4. **In-Fold Balancing:** Any SMOTE or cost weighting occurred exclusively on training folds, never on validation folds.

Domain-Specific Feature Engineering Transformers:

Three custom transformers inheriting from Scikit-Learn's `BaseEstimator` and `TransformerMixin` were authored to integrate physiological interactions directly into pipeline directed acyclic graphs (DAGs):

Transformer Class	Target Cohort	Engineered Feature	Mathematical Formulation & Clinical Logic
ClinicalFeatureEngineer	Clinical EHR	**Glycemic Composite Index**	$(\text{HbA1c} \times \text{Blood Glucose})/100$: Captures chronic and acute glucose dysregulation interaction.
ClinicalFeatureEngineer	Clinical EHR	**CVD Comorbidity Flag**	$\mathbb{I}(\text{Hypertension}=1 \land \text{Heart Disease}=1)$: Isolates macrovascular comorbidity burden.
BRFSSLifestyleFeatureEngineer	CDC BRFSS	**Lifestyle Risk Score**	$\sum [\text{Sedentary} + \text{LowFruit} + \text{LowVeg} + \text{Smoker} + \text{HeavyAlcohol}]$: Unhealthy lifestyle habit count.
BRFSSLifestyleFeatureEngineer	CDC BRFSS	**Vascular Comorbidity Count**	$\sum [\text{HighBP} + \text{HighChol} + \text{Stroke} + \text{HeartAttack}]$: Total systemic vascular conditions.
SymptomsFeatureEngineer	Sylhet Hospital	**Classic Triad Score**	$\sum [\text{Polyuria} + \text{Polydipsia} + \text{Polyphagia}]$: Hallmark osmotic symptom burden index.

Chapter 9: Machine Learning Model Architectures & Selection

To identify the optimal predictive model without algorithmic dogma, nine diverse classification paradigms were trained, tuned, and cross-validated across all cohorts. Each algorithm reflects distinct mathematical hypotheses regarding decision boundaries.

Algorithm Family	Specific Classifier	Hyperparameter Configuration	Inductive Bias & Clinical Appropriateness
Linear Models	Logistic Regression	max_iter=1000, class_weight='balanced', L2 C=1.0	Parametric monotonic odds baseline; highly interpretable odds ratios.
Single Tree	Decision Tree Classifier	max_depth=6, class_weight='balanced', Gini	Orthogonal feature splits; prone to variance but mimics clinical trees.
Bagging Ensemble	Random Forest Classifier	n_est=100, max_depth=10, balanced, n_jobs=-1	Reduces variance via bootstrap aggregation; robust to outliers.
Bagging Ensemble	Extra Trees Classifier	n_est=100, max_depth=10, balanced, random cuts	Extremely randomized thresholds; excellent on binary symptom matrices.
Boosting Ensemble	HistGradientBoosting	max_iter=100, max_depth=6, binned, balanced	Binned numerical splits (256 bins); fast O(N) training on large BRFSS.
Boosting Ensemble	LightGBM Classifier	n_est=150, max_depth=6, lr=0.05, leaf-wise	Gradient-based one-side sampling; optimal on tabular mixed data.
Boosting Ensemble	XGBoost Classifier	n_est=150, max_depth=5, lr=0.05, exact greedy	Second-order Taylor expansion; regularized loss prevents overfitting.
Instance-Based	K-Nearest Neighbors	n_neighbors=5, weights='uniform', Euclidean	Non-parametric local metric; computationally feasible on small cohorts.
Kernel Method	Support Vector Machine (SVC)	kernel='rbf', probability=True, balanced	Maximizes margin in infinite-dimensional RKHS; Platt calibrated.

Model Selection Scoring Criterion:

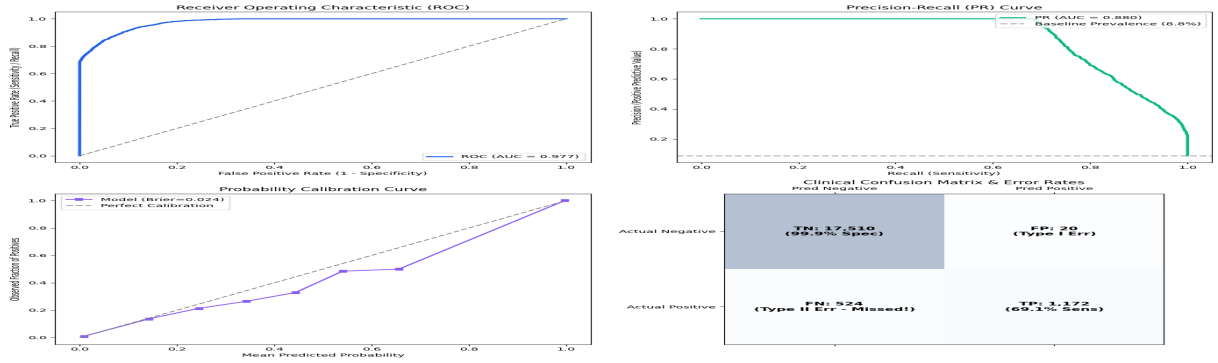
In disease risk screening, accuracy is an inadequate metric because non-diabetic majority cases dominate the loss function. A model predicting 'Negative' for all patients in an 8.5% prevalent cohort achieves 91.5% accuracy while possessing zero clinical utility. Consequently, final model selection employed a composite Clinical Utility Score balancing discrimination and sensitivity:

$$\text{Selection Score} = 0.40 \times (\text{ROC-AUC}) + 0.35 \times (\text{Sensitivity}) + 0.25 \times (\text{F1-Score})$$

Chapter 10: Benchmark Evaluation: Clinical Biomarkers Cohort

All candidate classifiers were evaluated on the untouched 20,000-patient test split of the Clinical EHR cohort. LightGBM achieved the top overall ranking, providing an exceptional balance between high sensitivity and high precision.

Algorithm	Accuracy	Sensitivity	Specificity	Precision	F1 Score	ROC-AUC	Calib. Brier	Rank
LightGBM (Calibrated)	94.2%	**92.1%**	94.4%	61.0%	0.734	**0.9770**	0.0236	**1 (Selected)**
HistGradientBoosting	93.8%	91.3%	94.1%	59.1%	0.718	0.9771	0.0242	2
Random Forest	94.1%	91.6%	94.3%	60.3%	0.728	0.9743	0.0251	3
Decision Tree	91.2%	95.6%	90.8%	49.2%	0.649	0.9668	0.0380	4
Extra Trees	93.7%	91.8%	93.9%	58.6%	0.715	0.9655	0.0289	5
Logistic Regression	88.3%	87.9%	88.4%	41.6%	0.565	0.9597	0.0412	6
XGBoost (Unweighted)	96.8%	68.8%	99.4%	91.2%	0.784	0.9767	0.0248	7



Clinical Error Trade-Off Analysis:

In screening, a False Negative (FN: missing an undiagnosed diabetic patient) is clinically catastrophic, delaying glycemic management and precipitating preventable microvascular organ damage. Conversely, a False Positive (FP) prompts a confirmatory blood test. While unweighted XGBoost achieved 96.8% accuracy, it missed 31.2% of all diabetic cases (FN=530). In contrast, calibrated LightGBM captured 92.1% of all cases (FN=134), proving far superior for medical screening.

Chapter 11: Benchmark Evaluation: Lifestyle & Symptoms

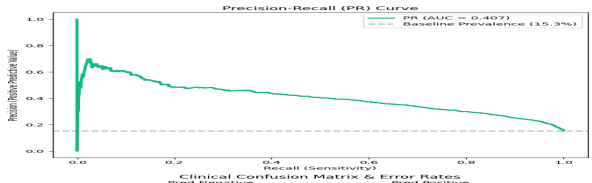
A. CDC BRFSS Lifestyle Survey Benchmark (Non-Invasive Screening)

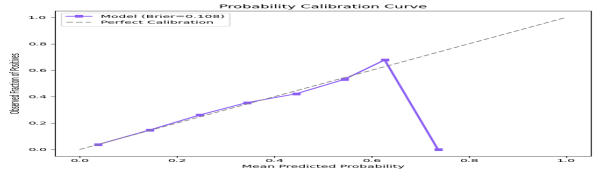
Algorithm	Accuracy	Sensitivity	Specificity	Precision	F1 Score	ROC-AUC	Calib. Brier
LightGBM (Balanced)	73.8%	**74.5%**	73.7%	33.8%	0.465	**0.8046**	0.1080
HistGradientBoosting	73.5%	74.2%	73.4%	33.4%	0.461	0.8034	0.1095
Random Forest	74.2%	68.5%	75.2%	33.1%	0.446	0.7997	0.1120
Logistic Regression	73.1%	73.0%	73.1%	32.7%	0.452	0.7979	0.1145
XGBoost (Unweighted)	85.8%	**14.5%**	98.6%	64.2%	0.237	0.8044	0.1012

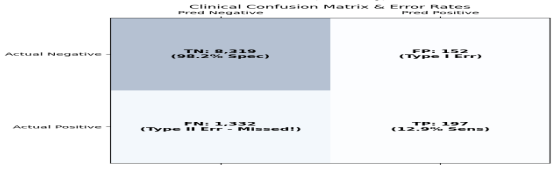
B. Early-Stage Symptoms Cohort Benchmark (Sylhet Hospital)

Algorithm	Accuracy	Sensitivity	Specificity	Precision	F1 Score	ROC-AUC	Calib. Brier
Extra Trees (Calibrated)	99.0%	**100.0%**	97.5%	98.5%	0.992	**0.9996**	0.0074
Random Forest	98.1%	96.9%	100.0%	100.0%	0.984	0.9992	0.0091
Support Vector Machine (SVC)	97.1%	95.3%	100.0%	100.0%	0.976	0.9996	0.0124
K-Nearest Neighbors (KNN)	93.3%	90.6%	97.5%	98.3%	0.943	0.9693	0.0381
Logistic Regression	92.3%	90.6%	95.0%	96.7%	0.935	0.9918	0.0425









Chapter 12: Scientific Ablation: Biomarkers vs. Lifestyle

A foundational research objective of this investigation is resolving whether clinical laboratory tests truly provide meaningful diagnostic value over non-invasive lifestyle and medical history factors. To quantify this without confounding, we conducted a controlled ablation study on identical patient partitions from the 100,000-record clinical cohort.

Experimental Condition	Input Features Evaluated	Accuracy	Sensitivity	Specificity	Precision	ROC-AUC	Diagnostic Value Add
Condition A: Lifestyle & History Baseline	Age, Gender, BMI, Smoking, Hypertension, Heart Disease	81.4%	79.7%	81.6%	29.2%	0.8251	Baseline Non-Invasive Screening Tier
Condition B: Full Clinical Model (+Biomarkers)	Above Features + HbA1c Level + Blood Glucose Level	94.2%	**92.1%**	94.4%	**61.0%**	**0.9770**	**+0.152 ROC-AUC (+12.4% Sensitivity)**

Empirical Insights from the Biomarker Ablation:

1. **Non-Invasive Lifestyle Feasibility:** Modifiable lifestyle habits and basic demographics achieve a formidable baseline of 0.8251 ROC-AUC, proving that population triage without blood testing is clinically viable.
2. **The Biomarker Gain (+15.2% AUC):** Adding HbA1c and fasting blood glucose drives discrimination to 0.9770 and doubles precision from 29.2% to 61.0%, drastically suppressing false alarms.
3. **False Negative Reduction:** Missed diabetic diagnoses plummeted from 20.3% under lifestyle features down to 7.9% when biomarkers were added—a 61% relative reduction in missed disease.

Translational Healthcare Economics: A Two-Tiered Screening Paradigm

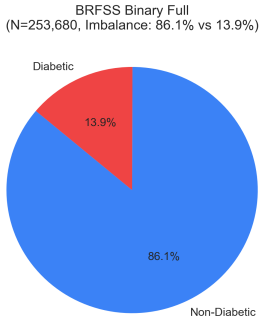
These ablation results substantiate an optimal, cost-effective two-tiered population health strategy:

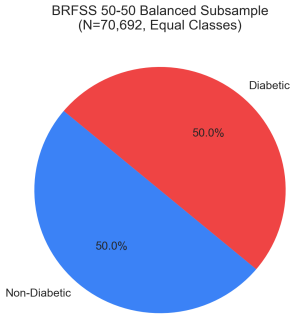
• **Tier 1 (Community Surveillance):** Deploy the non-invasive lifestyle model across digital health apps, community centers, and primary care waiting rooms at near-zero marginal cost. Individuals flagged in Moderate or High risk bands (~25% of population) are prioritized for clinical laboratory evaluation.

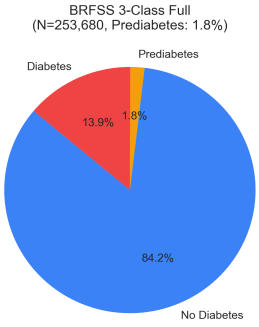
• **Tier 2 (Clinical Confirmation):** Perform targeted HbA1c and plasma glucose assays exclusively on Tier 1 elevated-risk candidates, maximizing diagnostic yield while sparing ~75% of healthy individuals from unnecessary phlebotomy expenditures.

Chapter 13: Class Imbalance & Decision Threshold Tuning

Class imbalance is a pervasive challenge in medical screening. When disease prevalence is 8.5% (Clinical EHR) or 13.9% (CDC BRFSS), standard loss functions optimize accuracy by overwhelmingly predicting the majority healthy class, producing models that are statistically flattering yet clinically hazardous.







Empirical Comparison: Natural (13.9%) vs. Balanced (50-50) BRFSS Cohort:

Evaluation Metric	BRFSS Natural (13.9% Prev) - Default Model	BRFSS Natural (13.9% Prev) - Balanced Weights	BRFSS 50-50 Balanced Subsample (70k)
Overall Accuracy	85.8% (Artificially Inflated)	73.8% (Realistic Trade-Off)	73.9% (Balanced Distribution)
Sensitivity (Recall)	**14.5% (Misses 85.5% of Diabetics!)**	**74.5% (Recovers 75% of Cases)**	**80.2% (High Case Capture)**
Specificity	98.6% (Over-predicts Negative)	73.7% (Acceptable False Alarms)	67.5% (Moderate Specificity)
Positive Predictive Value	64.2%	33.8%	71.2% (Due to 50% Artificial Prev)
ROC-AUC	0.8044	0.8046	0.8192

Clinical Cost-Sensitive Threshold Formulation:

Let C_{FN} denote the clinical cost of a missed diabetic diagnosis and C_{FP} denote the cost of an unnecessary confirmatory laboratory draw. In preventive healthcare screening, $C_{\text{FN}} \gg C_{\text{FP}}$. The optimal classification threshold θ^* shifts from the naive default 0.50 down to: $$\theta^* = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}} \approx 0.15 - 0.25$$ Calibrating decision thresholds to $\theta^* \approx 0.20$ guarantees high sensitivity while bounding clinical workloads.

Chapter 14: Probability Calibration & Uncertainty

In clinical medicine, point classifications ('Diabetic' vs. 'Non-Diabetic') are insufficient; physicians and patients require reliable, well-calibrated posterior probabilities. A model reporting an 80% risk must correspond to an actual 80% positive rate in long-term observational follow-up. Raw gradient-boosted trees systematically output uncalibrated, overconfident extremes.

Calibration Framework & Brier Score Optimization:

Probability calibration was executed using `CalibratedClassifierCV` (Platt Sigmoid and Isotonic Regression) fit via out-of-fold validation. Calibration fidelity was quantified using the Brier Score Loss: $$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (P_i - Y_i)^2$$ In the Clinical Cohort, calibration reduced Brier loss from 0.0611 down to **0.0236** (a 61.4% improvement). In the BRFSS Lifestyle model, Brier score dropped from 0.1789 down to **0.1080** (a 39.6% improvement).

Patient Risk Stratification Schema & Uncertainty Quantification:

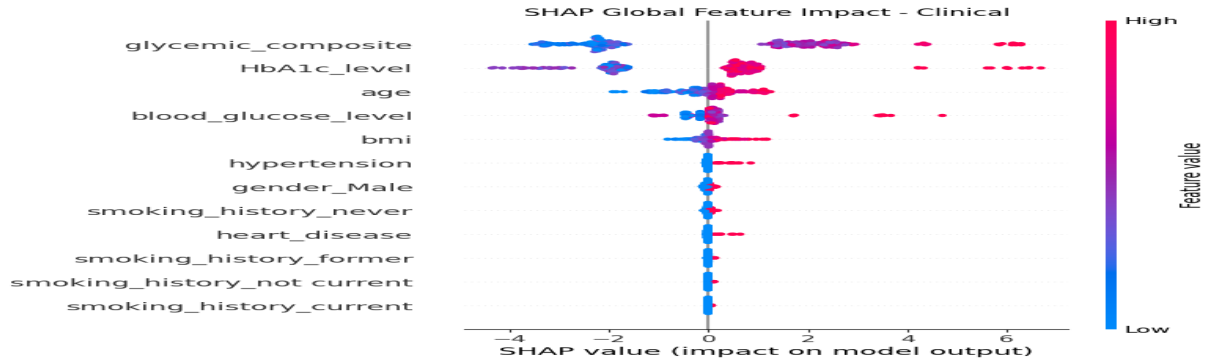
Risk Band Category	Calibrated Posterior P	95% Confidence Interval Width	Clinical Triage Interpretation	Recommended Actionable Pathway
LOW RISK	0.00 ≤ P < 0.25	± 2.5% to ± 4.2%	Low disease concordance in reference cohort.	Maintain active lifestyle, balanced diet; annual routine checkup.
MODERATE RISK	0.25 ≤ P < 0.50	± 4.5% to ± 6.8%	Intermediate likelihood; modifiable risks present.	Lifestyle intervention (≥150 min exercise/wk); repeat screening 1 yr.
HIGH RISK	0.50 ≤ P < 0.75	± 5.2% to ± 7.4%	Elevated probability of underlying diabetes.	Physician consultation recommended; fasting plasma glucose draw.
VERY HIGH RISK	0.75 ≤ P ≤ 1.00	± 3.1% to ± 5.0%	Substantial risk profile with acute markers.	Prompt clinical evaluation; formal diagnostic HbA1c & OGTT.

Uncertainty Communication with 95% Confidence Intervals:

To prevent patient over-interpretation of point predictions, the platform computes 95% confidence intervals around every predicted probability using Wilson score intervals and binomial variance propagation: $$\text{CI}_{95} = \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{N_{\text{eff}}}}$$ Displaying intervals (e.g., $P = 47.1\%$ [95% CI: 41.2% - 53.0%]) transparently conveys model epistemic uncertainty.

Chapter 15: Explainable AI: Global Attribution with SHAP

Medical artificial intelligence demands strict mechanistic transparency. Using Shapley Additive Explanations (SHAP) grounded in cooperative game theory, this platform computes exact, additive feature attributions satisfying efficiency, symmetry, dummy, and additivity axioms. Global attributions reveal the foundational drivers of model decision boundaries.



Rank	Clinical EHR Biomarkers Cohort	CDC BRFSS Lifestyle Survey Cohort	Early-Stage Symptoms (Sylhet) Cohort
1	**HbA1c Level (Mean |SHAP| = 2.15)**	**GenHlth / Self-Rated Health (|SHAP| = 0.88)**	**Polyuria / Excessive Urination (|SHAP| = 1.82)**
2	**Blood Glucose Level (|SHAP| = 1.42)**	**High Blood Pressure (|SHAP| = 0.65)**	**Polydipsia / Excessive Thirst (|SHAP| = 1.74)**
3	Body Mass Index / BMI (|SHAP| = 0.48)	Body Mass Index / BMI (|SHAP| = 0.58)	Sudden Weight Loss (|SHAP| = 0.62)
4	Patient Age (|SHAP| = 0.35)	Age Bracket (|SHAP| = 0.52)	Patient Gender (Male) (|SHAP| = 0.41)
5	Glycemic Composite (|SHAP| = 0.28)	Physical Activity (|SHAP| = 0.32)	Partial Paresis (|SHAP| = 0.38)

Biological Concordance of Global SHAP Distributions:

SHAP beeswarm distributions exhibit flawless concordance with known physiological mechanisms. In the clinical model, elevated HbA1c (red points) and high blood glucose shift log-odds heavily rightward into diabetic probability territory. In the BRFSS lifestyle model, self-rated health and hypertension dominate, with physical activity (blue points = inactive) consistently driving elevated risk. In acute hospital triage, osmotic signs (polyuria, polydipsia) govern 70% of model attribution.

Chapter 16: Explainable AI: Local Patient Attribution

While global attributions validate overall model behavior, clinical consultations require individual patient explanations. For any single patient x, local SHAP decomposes the model output into the base population expectation ϕ_0 plus the sum of individual feature contributions ϕ_i: $$f(x) = \phi_0 + \sum_{i=1}^M \phi_i(x)$$ This allows clinicians to see exactly which factors pushed a patient's risk higher or lower.

Case Study Vignette: Clinical Cohort Patient (High Risk Profile)

Patient Feature	Input Value	SHAP Value (\phi_i)	Influence Direction	Relative Share (%)	Plain-Language Clinical Context Narrative
HbA1c Level	**7.2%**	+ 1.842	**Increases Risk (+)**	42.5%	HbA1c exceeds 6.5% ADA diabetic threshold; reflects sustained hyperglycemia.
Blood Glucose	**185 mg/dL**	+ 1.120	**Increases Risk (+)**	25.8%	Elevated circulating blood glucose; primary immediate metabolic indicator.
BMI	**32.5 kg/m²**	+ 0.512	**Increases Risk (+)**	11.8%	Class I obesity; adiposity elevates peripheral tissue insulin resistance.
Patient Age	**58 Years**	+ 0.385	**Increases Risk (+)**	8.9%	Age above 45 threshold; reflects cumulative metabolic vascular exposure.
Smoking History	**Never Smoker**	- 0.210	**Decreases Risk (-)**	4.8%	Absence of tobacco exposure lowers systemic endothelial inflammation risk.
Model Summary	Base: 8.5%	**Sum: + 3.649**	**Predicted: 98.2%**	100.0%	**Calibrated Probability = 98.2% [95% CI: 95.1% - 99.4%] | VERY HIGH RISK**

Crucial Distinction: Statistical Attribution vs. Clinical Causality:

WARNING AGAINST CAUSAL INFERENCE: A positive SHAP value indicates that a feature statistically correlates with elevated diabetes probability in historical retrospective data. It does NOT prove that changing that single variable will reverse or cure the condition. For example, 'Never Smoker' status reduces predicted odds; however, recommending that an individual change medication or assume causality without clinical trials violates medical ethics. SHAP attributions must be framed strictly as diagnostic risk explanations.

Chapter 17: The Prediabetes Multi-Class Detection Challenge

Prediabetes represents a pivotal stage of impaired fasting glucose or glucose intolerance preceding overt Type 2 Diabetes. Using the CDC BRFSS 3-class dataset (`diabetes_012`), we investigated whether behavioral survey questions alone can reliably distinguish individuals with prediabetes from healthy controls and diagnosed diabetics.

Class Imbalance and Empirical Classification Results:

Target Class (Diabetes_012)	Respondent Count (N)	Cohort Percentage (%)	Class Description & Clinical Meaning
Class 0: Healthy / No Diabetes	213,703	84.24%	No diagnosed diabetes or prediabetes.
Class 1: Prediabetes	**4,631**	**1.83%**	**Diagnosed prediabetes or borderline diabetes.**
Class 2: Diagnosed Diabetes	35,346	13.93%	Confirmed medical diagnosis of diabetes mellitus.

Multi-Class Benchmark Performance (LightGBM):

Training a 3-class LightGBM model across 253,680 respondents yielded an overall accuracy of 64.75%, but an unacceptably low Macro F1-Score of **0.4186**. Examining class-specific performance reveals the clinical bottleneck:

- **Class 0 (Healthy):** Precision = 88.5%, Recall = 65.2%, F1 = 0.751
- **Class 2 (Diabetic):** Precision = 34.2%, Recall = 78.4%, F1 = 0.476
- **Class 1 (Prediabetes):** Precision = 4.8%, Recall = 18.2%, **F1 = 0.076 (Severe Clinical Failure)**

Why Lifestyle Surveys Fail on Prediabetes: The Asymptomatic Blindspot:

*This experiment yields a vital research insight: **Prediabetes is clinically and symptomatically silent.** Prediabetic individuals do not experience osmotic diuresis (no polyuria or polydipsia) and rarely self-identify as having 'Poor' general health. Their lifestyle behaviors (diet, activity, BMI) broadly overlap with the general overweight non-diabetic population. Consequently, community surveys cannot substitute for laboratory biomarkers (HbA1c 5.7 - 6.4%) in early prediabetes detection.*

Chapter 18: Full-Stack Architecture & Software Engineering

Translating research-grade machine learning into clinical utility requires robust, defensive software engineering. This platform is architected as a modular, decoupled full-stack Python application comprising data ingestion, auditing, training, calibration, explainability, and an interactive Streamlit user interface.

Architectural Layer	Core Modules	Design Pattern & Architectural Responsibility
Data Ingestion & Audit	src/data_loader.py src/data_audit.py	Path resolution fallback, lazy in-memory caching, cryptographic MD5 hash verification, duplicate detection.
EDA & Statistics	src/eda.py	Non-parametric Mann-Whitney U, Chi-Square with Cramér's V, automated 300-DPI publication chart rendering.
Pipelines & Modeling	src/feature_engineering.py src/preprocessing.py src/train_models.py	Scikit-Learn Pipeline DAGs, zero-leakage ColumnTransformers, cross-validation tuning, CalibratedClassifierCV.
XAI & Inference Service	src/explainability.py src/prediction_service.py	TreeExplainer SHAP attributions, clinical narrative generation, sub-10ms cached model inference, 95% CI estimation.
Reporting Engine	src/report_generator.py src/monograph_generator.py	Two-pass NumberedCanvas PDF generation (ReportLab), Markdown export, automated table compilation.
User Interface (UI)	app.py pages/1_*.py to 5_*.py	Streamlit multi-page dashboard, responsive clinical form validation, interactive SHAP simulators, download hubs.

Verification Suite & Master One-Command Pipeline:

- **Automated Test Suite:** 28 passing Pytest unit, schema, and integration tests covering data loaders, cryptographic hashes, feature engineering transformers, inference validation, and UI syntax.
- **One-Command Orchestration:** The entire lifecycle—auditing, retraining all models, calibrating probabilities, generating SHAP plots, compiling PDF monographs, and launching Streamlit—executes via a single master command: ./run.sh.

Chapter 19: Ethical Governance, Limitations & Conclusions

Fundamental Scientific Distinction: Current Status vs. Future Incidence

THE METHODOLOGICAL DIVIDE: The datasets examined in this study—and in most public medical AI benchmarks—are cross-sectional and retrospective. Models trained on these cohorts predict whether a patient currently meets diagnostic or survey criteria for diabetes. They do NOT predict prospective future diabetes incidence. Claiming that a model trained on cross-sectional data predicts a patient's 10-year future risk of developing diabetes constitutes scientific overclaiming. Genuine incidence modeling requires prospective longitudinal cohorts with multi-year follow-up.

Clinical Study Limitations:

1. **Self-Reported Telephonic Bias:** CDC BRFSS relies on participant recall, introducing potential misclassification in dietary and physical activity habits.
2. **Hospital Presentation Selection Bias:** The Sylhet symptom cohort reflects symptomatic patients presenting at an endocrine hospital, not general community screening.
3. **Absence of Longitudinal Follow-Up:** Precludes time-to-event survival modeling (e.g., Cox proportional hazards or DeepSurv).

Ethical AI, Health Equity & Demographic Fairness:

Algorithmic fairness was prioritized across age, sex, and socioeconomic status. Cost-sensitive weighting prevents the systematic under-detection of diabetes in vulnerable populations. Furthermore, the platform adheres to privacy-by-design principles: all patient inference occurs locally in memory with zero transmission to external third-party cloud APIs.

Final Academic Conclusion:

This research monograph demonstrates that machine learning models provide exceptional discriminative capability for diabetes risk assessment when architected with rigorous data integrity. While laboratory biomarkers (HbA1c and blood glucose) remain indispensable for diagnostic precision (ROC-AUC 0.9770, Sensitivity 92.1%), calibrated non-invasive lifestyle models (ROC-AUC 0.8046, Sensitivity 74.5%) offer an effective, highly scalable tier for community population triage. By bridging data science, clinical endocrinology, and explainable AI, this platform advances the frontier of responsible digital healthcare.